

Audrey Shih

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EDUCATION

Stanford University	Stanford, CA
Ph.D., Chemical Engineering, GPA 3.95	Expected June 2025
M.S., Chemical Engineering	2020 – 2023
Princeton University	Princeton, NJ
B.S.E., Chemical and Biological Engineering, <i>cum laude</i>	2016 – 2020
Minor, Materials Science and Engineering	

EXPERIENCE

Stanford University , Stanford, CA	2021 – Present
<i>Doctoral Researcher</i> PI: Gerald G. Fuller	
- Developed and optimized a portable magnetic stress rheometer using CAD-designed components and Raspberry Pi, reducing device costs from ~\$90k to \$160 for use in medical settings. Applied hands-on engineering, testing, and troubleshooting to enhance real-time fluid flow predictions for abscess drainage and biopolymeric materials. - Engineered 3D bioprinting systems for rheological analysis of bioinks in diffusion-driven crosslinking, solving technical challenges related to optimizing mechanical properties of printed complex biological tissues. - Managed lab safety protocols and compliance as designated lab safety officer, attending quarterly environmental health and safety meetings with University officials to ensure regulatory adherence in research environments. - Instructed and mentored both undergrad and grad students in <i>Mechanics of Soft Matter: Rheology (CHEMENG 470)</i>.	
Stanford University , Stanford, CA	2021
<i>Doctoral Researcher</i> PI: Joseph M. DeSimone	
- Developed and optimized high-resolution 3D Continuous Liquid Interface Production (CLIP) printing through experiments, SEM imaging, and simulations of optics and photopolymerization transport and kinetics. - Utilized CAD and 3D-slicing software to test and optimize 3D printer settings, enhancing resolution and precision in additive manufacturing processes.	
Princeton University , Princeton, NJ	2018 – 2020
<i>Undergraduate Researcher</i> PI: Sujit S. Datta	
- Investigated elastic turbulence in polymer flow by CAD designing 3D-printed microfluidic model porous channels. - Featured in ACEE's first ever spotlight article on undergraduate contributions to environmental studies. - Won Outstanding Thesis Award, thesis featured in profile article by the School of Engineering and Applied Science.	
TU-Dortmund, Fakultät Bio und Chemieingenieurwesen , Dortmund, Germany	2018
<i>Research Intern</i> Armin Quentmeier	
- Commandeered the Biochemical and Chemical Engineering department's contribution to the 2018 Engineering Meets Art exhibition, designed project involving photographs of microorganisms and colonies. - Wrote 40-page manual of protocols for inoculation and dilution techniques, instructions for using cameras, lenses, microscopes, and Adobe Photoshop Elements, and analyzed microscopic images using various software. - Final projects displayed at exhibition from May – June 2019.	
Princeton University , Princeton, NJ	2017 – 2018
<i>Undergraduate Researcher</i> PI: Celeste M. Nelson	
- Analyzed role of tissue mechanics in breast cancer tumor development using immunostaining techniques. - Synthesized and crosslinked polyacrylamide gels, passaged and stained cells, analyzed results with mass spectrometry, Western blot, and confocal microscopy.	

SKILLS

Technical/Laboratory: Fluid mechanics; rheology; polymers; CAD (Fusion 360, Onshape); 3D printing (SLA, FDM, CLIP); hardware/embedded systems (PCB assembly, Raspberry Pi); SEM; GPC; DMA; PCR; cell culture; immunostaining; confocal microscopy imaging

Software: Java, MATLAB, Python; ImageJ; TRIOS; PIV; Adobe Illustrator, Photoshop; LaTeX; Microsoft Office

PUBLICATIONS & PATENTS

- **Shih, A.**, Chung, S. J., Brunel, L. G., Christakopoulos, F., Heilshorn, S. C., Fuller, G. G., *In situ* rheological measurements of bioink crosslinking. *In preparation*.
- **Shih, A.**, Chung, S. J., Shende, O. B., Herwald, S. E., Vezeridis, A. M., Fuller, G. G., [Viscoelastic measurements of abscess fluids using a magnetic stress rheometer](#). *Physics of Fluids* 36 (11): 111919 (2024), *Cover Article*.
- Cai, P. C., Braunreuther, M., **Shih, A.**, Spakowitz, A. J., Fuller, G. G., Heilshorn, S. C., [Air-liquid intestinal cell culture allows in situ rheological characterization of intestinal mucus](#). *APL Bioengineering* 8 (2): 026112 (2024).
- DeSimone, J. M., Jacobson, G. B., Dulay, M. T., Lee, B. J., Hsiao, K., Rajesh, N., Driskill, M. M., **Shih, A.**, et. al., *Polymeric microstructures and systems and methods for making same*. [Patent number WO2023049267A1 \(2023\)](#).
- Hsiao, K., Lee, B. J., Samuelsen, T., Lipkowitz, G., Kronenfeld, J. M., Ilyn, D., **Shih, A.**, et al., [Single-digit-micrometer-resolution continuous liquid interface production](#). *Science Advances* 8, eabq2846 (2022).
- Browne, C. A., **Shih, A.**, Datta, S. S., [Bistability in the Unstable Flow of Polymer Solutions in Porous Media](#). *Journal of Fluid Mechanics* 890:A2 (2020).
- Browne, C. A., **Shih, A.**, Datta, S. S., [Pore-Scale Flow Characterization of Polymer Solutions in Microfluidic Porous Media](#). *Small* 16, 1903944 (2019).

SELECTED AWARDS

2024	Excellence in Fluid Mechanics Research and Oral Presentation Award, AIChE Annual Meeting
2024, 2020	Sigma Xi Scientific Research Honor Society Nominee
2023	Chemical Engineering Department Service Leadership Award
2021	Judges' Vote and Audience Choice poster awards, Stanford MIPS Retreat, Woodside, CA
2020	National Science Foundation (NSF) Graduate Research Fellowship
2020	Lore von Jaskowsky Memorial Prize, Princeton School of Engineering and Applied Sciences
2020	Materials Science and Engineering Department Outstanding Senior Thesis Award
2020	Sigma Xi Scientific Research Honor Society Nominee
2019	PSEG Best Poster Award, ACEE Annual Meeting, Princeton, NJ

CONFERENCE TALKS

- **Shih, A.**, Chung, S. J., Shende, O. B., Brunel, L. G., Christakopoulos, F., Herwald, S. E., Vezeridis, A. M., Fuller, G. G., *Magnetic stress rheometer for biological fluid characterization*. American Society of Chemical Engineers Annual Meeting (AIChE 2024), San Diego, CA, Oct. 2024.
- **Shih, A.**, Chung, S. J., Shende, O. B., Herwald, S. E., Vezeridis, A. M., Fuller, G. G., *Rheological Characterization of Biological Fluids*. NETZSCH Introduction to Rheology and Thermal Analysis Workshop, Palo Alto, CA, Oct. 2024.
- **Shih, A.**, Chung, S. J., Herwald, S. E., Vezeridis, A. M., Fuller, G. G., *Magnetic stress rheometer for biological fluid characterization*. American Society of Chemical Engineers Annual Meeting (AIChE 2023), Orlando, FL, Nov. 2023.
- **Shih, A.**, Chung, S. J., Herwald, S. E., Vezeridis, A. M., Fuller, G. G., *Magnetic stress rheometer for biological fluid characterization*. XIXth International Congress on Rheology (ICR 2023), Athens, Greece, Aug. 2023.
- **Shih, A.**, Chung, S. J., Herwald, S. E., Vezeridis, A. M., Fuller, G. G., *Magnetic stress rheometer for biological fluid characterization*. Society of Rheology Annual Meeting (SOR 2022), Chicago, IL, Oct. 2022.

LEADERSHIP & MENTORSHIP

Stanford University , Stanford, CA	2022 – Present
<i>Research Mentor</i>	

- Mentored two undergraduate students (Stanford University, Shenzhen University) and one high school intern (Germantown Friends School), teaching CAD, Python, MATLAB, ImageJ, fluid mechanics, and polymer theory.
- Provided hands-on guidance in hardware, coding, and data analysis, fostering technical skills and problem-solving abilities for real-world research projects in rheology, medicine, and 3D bioprinting.

Graduate Life Office , Stanford University , Stanford, CA	2021 – Present
<i>Community Associate</i>	

- Coordinated 12 events annually to foster community among graduate student population, managing a \$3,600 budget for diverse and inclusive programs, including welcome, orientation, and multicultural initiatives.
- Advised graduate residents on support services and professional development, utilizing annual training on suicide prevention, servant leadership, bias intervention, and sexual harassment prevention.

Self-employed

2018 – present

STEM tutor and clarinet instructor

- Taught math and chemistry to students from ages 10-17 and clarinet performance to adult beginner students.

Graduate Student Action Committee, Stanford University, Stanford, CA

2022 – 2023

Social Events Chair | Department of Chemical Engineering

- Served as liaison between ChemE student community and administration through quarterly meetings.
- Spearheaded multi-departmental events to expand the department's community, promoting social and collaborative connections across various department programs.

Stanford University, Stanford, CA

2022 – 2023

First-Year Mentor | Department of Chemical Engineering

- Guided first-year students in selecting labs, managing coursework, and preparing for qualification exams, sharing resources and support through three meetings per quarter.

Stanford Splash, Stanford, CA

2023

Teacher Volunteer

- Analyzed role of tissue mechanics in breast cancer tumor development using immunostaining techniques.
- Taught hands-on workshop on polymer crosslinking to middle- and high-school students in two classes.

PROFESSIONAL AFFILIATIONS

Society of Rheology, American Institute of Chemical Engineers